SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – COMPARITIVE ANALYSIS TASK (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO3	**Assessment:**	Assessment 3: Comparitive Analysis Task
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.***		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct
I T Bhupal/192425243 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: bhupalreddy.

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Comparitive Analysis Task

1. CNN vs Traditional Neural Networks
Compare Convolutional Neural Networks (CNNs) with traditional Artificial Neural Networks (ANNs) for image classification. Analyze them based on architecture, feature extraction, number of parameters, spatial information, computational requirements, and suitability for image-based applications.

2. Convolution Layer vs Fully Connected Layer

Compare the Convolution Layer and Fully Connected Layer in a CNN. Discuss their differences in terms of connectivity, parameter usage, spatial information, feature extraction, and their role in the overall CNN architecture. Explain why both layers may be required for image classification.

3. Max Pooling vs Average Pooling

Perform a comparative analysis of Max Pooling and Average Pooling. Explain their working principles, advantages, limitations, and effect on feature maps. Give suitable real-world situations where Max Pooling would be preferred over Average Pooling and vice versa.

4. Different Filters/Kernels in CNN

Compare the use of different convolution filters/kernels for feature extraction. Explain how kernels can detect edges, textures, curves, and complex patterns. Analyze why multiple filters are required in a CNN and how the learned features differ between shallow and deeper layers.

5. Parameter Sharing vs Fully Connected Processing

Compare parameter sharing in CNNs with the conventional approach of assigning separate weights to different input locations. Analyze how parameter sharing affects the number of trainable parameters, computational efficiency, and the ability of CNNs to recognize objects at different locations in an image.

6. CNN With and Without Regularization

Consider two CNN models trained for the same image-classification task. Model A uses no regularization, while Model B uses Dropout, Data Augmentation, and L2 regularization.
Compare the two approaches and analyze:

- Training performance
- Validation performance
- Overfitting
- Generalization ability
- Model complexity

Conclude which approach would be more reliable for real-world image classification and why.

7. AlexNet vs ResNet

Conduct a detailed comparative analysis of AlexNet and ResNet based on:

- Architecture
- Network depth
- Convolutional layers
- Training difficulty
- Number of parameters
- Vanishing-gradient problem
- Skip/residual connections
- Performance on complex image-classification tasks

Finally, recommend which architecture would be more suitable for a modern image-classification application and justify your answer.

8. Shallow CNN vs Deep CNN

Compare a shallow CNN containing a few convolutional layers with a deep CNN containing many convolutional layers. Analyze their ability to learn low-level and high-level features, computational requirements, training complexity, overfitting risk, and performance. Explain why deeper architectures such as ResNet are useful for complex image-recognition problems.

9. CNN Architecture Comparison for Different Applications

Consider the following applications:

- Face Recognition
- Medical Image Classification
- Autonomous Vehicles
- Handwritten Digit Recognition

Analyze whether the same CNN architecture should be used for all four applications. Compare the requirements of these applications and explain what factors should be considered while selecting a CNN architecture, such as dataset size, image complexity, computational resources, accuracy, and inference speed.

10. Complete CNN Architecture Analysis

Consider a CNN architecture consisting of:

Input Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Fully Connected Layer → Output

Perform a complete analysis of this architecture. Explain the purpose of each component, how information changes as it moves through the network, and how the architecture finally produces a classification result.

Compare this architecture with a ResNet-based architecture and explain how residual/skip connections can improve the training of deeper networks.

1. CNN vs Traditional Neural Networks (ANN)

Comparative Analysis for Image Classification:

- **Architecture:** CNNs process data in its original 2D grid structure (height × width). Traditional ANNs require the image to be immediately "flattened" into a single 1D vector (a long line of pixels).

- **Spatial Information:** Because ANNs flatten the image, they destroy the critical spatial relationships between neighboring pixels. CNNs preserve these spatial relationships, allowing them to understand shapes and local structures.

- **Feature Extraction:** CNNs perform automatic feature extraction using convolution layers, learning what edges and textures look like on their own. Traditional ANNs cannot do this effectively and often require human engineers to manually extract features beforehand.

- **Number of Parameters:** ANNs connect every single pixel to every single neuron, resulting in a massive, memory-heavy number of parameters. CNNs use "parameter sharing" (reusing the same small filter across the whole image), which drastically reduces the parameter count.

- **Computational Requirements:** Due to the dense connections, ANNs are computationally expensive and prone to overfitting when applied to high-resolution images. CNNs are highly computationally efficient and scalable.

- **Suitability:** CNNs are the industry standard for image-based applications (vision, medical imaging, object detection), while ANNs are better suited for tabular or structured numerical data.

2. Convolution Layer vs Fully Connected Layer

Differences and Roles in the CNN Architecture:

- **Connectivity:** The Convolution Layer uses *local connectivity*—it only looks at a small patch of the image at a time using a filter. The Fully Connected (FC) Layer uses *global connectivity*—every neuron is connected to every single output from the previous layer.

- **Parameter Usage:** Convolution layers have very few parameters because they share the same filter weights across the image. FC layers contain the vast majority of the network's parameters, as every connection requires a unique weight.

- **Spatial Information:** Convolution layers preserve 2D spatial dimensions (height and width). Before entering the FC layer, the data is flattened into 1D, meaning all spatial location data is permanently lost.

- **Feature Extraction vs. Classification:** The Convolution layer acts as the **feature extractor**, identifying *where* and *what* the patterns (edges, textures) are. The FC layer acts as the **classifier**, looking at the combination of all extracted features to make a final decision (e.g., "If whiskers and pointy ears are present, it is a cat").

Why Both Are Required: A network cannot classify an image without first understanding its visual contents, and it cannot output a final label using just scattered 2D pattern maps. The convolution layers build the visual understanding, and the FC layer translates that understanding into a final mathematical prediction.

3. Max Pooling vs Average Pooling

Comparative Analysis:

- **Working Principles:** Max Pooling slides a window across a feature map and keeps only the *highest* pixel value in that region. Average Pooling calculates the *arithmetic mean* of all pixel values within that window.

- **Effect on Feature Maps:** Max Pooling acts as a signal amplifier, aggressively highlighting the most prominent features and discarding background noise. Average Pooling acts as a smoothing agent, diluting strong signals but retaining the overall background context.

- **Advantages & Limitations:** Max pooling is excellent at extracting sharp, distinct features but loses background information. Average pooling preserves the global context but can blur or weaken the most important features.

Real-World Situations:

- **When to use Max Pooling:** Preferred in applications where distinct, sharp features matter most. Examples include object detection (finding sharp boundaries of cars) or medical X-ray diagnosis (identifying bright, hyperdense tumor spots against dark tissue).

- **When to use Average Pooling:** Preferred when the overall context of the image is more important than specific sharp edges. Examples include identifying the overall scenery (e.g., classifying an image as a "beach" based on a smooth blend of sand and sky) or in the very final layers of modern deep networks (Global Average Pooling) to prepare data for classification.

4. Different Filters/Kernels in CNN

Feature Extraction and Multiple Filters:

- **How Kernels Detect Patterns:** A kernel is a small grid of numbers. As it slides over the image, it performs mathematical dot products with the pixels beneath it. If the pixels match the exact numerical pattern of the kernel, the output is a high number (an "activation"). This allows specific kernels to act as dedicated edge detectors, texture scanners, or curve finders.

- **Why Multiple Filters Are Required:** A single image contains thousands of overlapping details. One filter can only look for one specific thing (e.g., vertical lines). By using multiple filters in a single layer (often 32, 64, or 128), the network can scan for horizontal lines, diagonal edges, color gradients, and textures all at the exact same time.

- **Shallow vs. Deeper Layers:**
  - *Shallow Layers (early in the network):* Filters act as low-level detectors. They find simple, fundamental patterns like sharp edges, straight lines, and basic color contrasts.
  - *Deep Layers (later in the network):* Filters combine the simple lines and edges from the shallow layers to detect high-level, complex semantic features—such as the shape of a wheel, the texture of fur, or an entire human eye.

5. Parameter Sharing vs Fully Connected Processing

Comparison and Impact on CNNs:

- **The Concept:** In Fully Connected processing, every single pixel location on the screen is assigned its own unique, independent mathematical weight. In Parameter Sharing, a CNN takes a single set of weights (e.g., a 3×3 kernel containing just 9 weights) and slides it across the entire image.
- **Number of Trainable Parameters:** Parameter sharing drastically reduces the size of the model. Instead of needing 1,000,000 separate weights to scan a 1000×1000 image, a CNN only needs 9 shared weights for a single filter.
- **Computational Efficiency:** Because there are millions fewer parameters to calculate and store in memory, parameter sharing makes CNNs incredibly fast, lightweight, and capable of running on standard hardware.
- **Object Location (Translation Invariance):** Fully connected networks learn patterns based on exact pixel locations—if a cat moves from the center of the image to the top left, the network fails to recognize it. Because parameter sharing slides the *same* feature detector over the *entire* image, a CNN will successfully detect the cat regardless of where it appears in the frame.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________